Long-Life Diamond Carbon Skeleton Particle Electrode Technical Whitepaper

Published: January 2026 | Topic: Next-gen industrial wastewater 3D electrolysis core consumable derived from solid-state battery technology

1. Executive Summary

In high-concentration refractory organic wastewater and precious metal recovery, traditional fillers face "short life, high energy consumption, single function." **Nanjing CuFeng Mechanical & Electrical Technology Co., Ltd.** has developed a novel composite particle electrode **with micron-scale diamond as structural skeleton and highly conductive carbon as matrix**, based on advanced interface engineering principles.

The product well integrates diamond's **high mechanical durability** and carbon's **excellent electrochemical activity**. It not only offers a lower life-cycle-cost solution for dyeing and pharmaceutical chemical wastewater, but also demonstrates unique resource recovery value in **gold refining and nuclear industry**.

2. Technical Principles & Core Advantages

The core technology originates from **high-modulus anode host materials for all-solid-state lithium metal batteries**. We apply the same "rigid yet flexible" microstructure design to solve engineering challenges in water treatment.

2.1 Biomimetic Rigid Conductive Composite

- ✓ **Continuous conductive phase (electron channel):** Highly graphitized carbon as matrix tightly wraps and connects diamond aggregate. Special interface modification and densification achieve strong metallurgical bonding between hard aggregate and conductive matrix, ensuring the material does not disintegrate or delaminate under long-term water flow.
- ✓ **Ultra-hard aggregate phase (wear core):** Industrial diamond powder is introduced. Diamond's high hardness (Mohs 10) builds a dense rigid network inside the particle, bearing all mechanical collision and friction during fluidization. **Result: Compressive strength > 100 MPa, achieving "zero sludge" operation.**

2.2 BDD-Like Advanced Oxidation Mechanism

Diamond phase has a wide electrochemical window. Under electric field, diamond surface efficiently generates **hydroxyl radicals (·OH)** with oxidation potential up to 2.8V. Compared with traditional adsorption-dominated materials, this solution mineralizes refractory organics indiscriminately; COD removal rate is 20–30% higher than traditional materials.

3. Core Performance Benchmark

Data below is based on standard 3D electrolysis reactor test results, demonstrating the significant advantages of this solution in engineering applications.

Dimension	Recommended Diamond Carbon Skeleton (This Solution)	
Mechanical Strength (Compression/Wear)	Recommended > 100 MPa (High, zero pulverization)	(Ea
Service Life	Recommended 5–8 years	3
Conduction Mode	Recommended Bulk conductive (Wear does not fail)	Bul
Catalytic Mechanism	Recommended Advanced oxidation (·OH) Efficient mineralization	Physi Eas
Hazardous Waste	Recommended None	Large wa (High
Energy Consumption	Recommended Low (low cell voltage)	

4. Target Market Application Map

This product is suitable not only for traditional wastewater treatment, but also has unique advantages in resource recovery and specialty fields.

🏆 Gold & Precious Metal Smelting

High margin Resource recovery

Core pain: Cyanide highly toxic and hard to degrade; trace gold loss.

Value: 1. ·OH efficiently oxidizes and breaks cyanide. 2. Porous conductive skeleton enables in-situ electrodeposition for gold recovery.

🏭 Plating/PCB Park Wastewater

Heavy metals Complexes

Core pain: Heavy metal complexes hard to break; COD/ammonia nitrogen hard to meet standards.

Value: Strong oxidation instantly breaks complexes; releases metal ions for cathode deposition and recovery (e.g., Cu, Ni); achieves compliant discharge.

🧪 Pharmaceutical/Pesticide Chemical Wastewater

High salt High toxicity

Core pain: High salt inhibits biochemistry; antibiotics/aromatics highly toxic.

Value: Diamond skeleton resists high-salt corrosion; strong oxidation directly opens rings and breaks chains; significantly improves biodegradability.

👕 Dyeing/Paper Wastewater Upgrading

Large volume Cost reduction

Core pain: Activated carbon frequently pulverizes; sludge disposal cost high.

Value: Direct replacement of existing activated carbon filler; no replacement for 5 years; annual operating cost reduced by 40%+.

☢️ Nuclear Industry Wastewater

High barrier Long life

Core pain: Maintenance difficult; equipment must be highly stable; minimize consumable replacement.

Value: Diamond highly radiation-resistant; long-life, maintenance-free characteristics well match nuclear industry "minimal/unmanned" requirements.

🛢️ Oilfield Injection Water/Fracturing Fluid

Oil-resistant Non-compacting

Core pain: Oil and polymer content; filler easily compacts and blocks.

Value: Chemically inert surface, low oil adsorption; strong oxidation directly degrades polymers; prevents formation blockage.

5. Economics & Cooperation Models

💰 **Cost reduction:** Although unit purchase price is higher than activated carbon, long life (>5 years vs 0.5 years) and no hazardous waste disposal cost reduce life-cycle cost by **50%+**.

🤝 **Flexible models:**

1. **Direct sales:** For new projects or major replacement.

2. **Leasing:** Annual fee for operators; no asset pressure.
3. **Revenue share:** For gold recovery projects; share of recovered metal value.

Start Cooperation

Diamond carbon skeleton particle electrode — turning industrial wastewater treatment from "cost center" to "value center."

Contact us for samples or on-site pilot scheduling.

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